

IE598 Machine Learning in Finance Lab

Module 6 Homework (Cross validation)

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1 Introduction

Using the `ccdefault.csv` dataset, the data were divided into 90% training data and 10% test data using stratified sampling. The decision tree model was evaluated using repeated random train-test splits (Monte Carlo Cross Validation) and 10-fold cross-validation. Since a single train-test split may give an estimate that depends on the particular observations selected for the training and test sets, these two methods are used to obtain more reliable estimates of the model's performance on unseen data.

In Monte Carlo Cross Validation, the training data are repeatedly divided into training and validation sets using different random splits. In contrast, 10-fold cross-validation divides the training data into 10 folds and uses each fold once as the validation set while the remaining folds are used for training. By evaluating the model across multiple different samples, we can assess both its average accuracy and the variability of its performance. The resulting estimates are then compared with the accuracy obtained from the original 10% holdout test set, which is kept separate from the two procedures.

2 Monte Carlo Cross Validation

The original 90% training set was repeatedly divided into 90% training and 10% validation sets using ten different random states. For each split, in-sample and out-of-sample accuracy were calculated, see Table 1.

Table 1: Monte Carlo Cross Validation Results

Random State	In-Sample Accuracy	Out-of-Sample Accuracy
100	0.9994	0.7222
101	0.9994	0.7393
102	0.9994	0.7278
103	0.9994	0.7189
104	0.9995	0.7178
105	0.9994	0.7230
106	0.9994	0.7204
107	0.9995	0.7311
108	0.9994	0.7048
109	0.9994	0.7259
Mean Accuracy	0.9994	0.7231
Std. Dev.	0.0001	0.0091

The decision tree achieved a mean in-sample accuracy of 0.9994 with a standard deviation of 0.0001. The mean out-of-sample accuracy was 0.7231 with a standard deviation of 0.0091. The difference between the in-sample and out-of-sample accuracy indicates that the decision tree fits the training data well but has lower performance on unseen data.

3 Cross Validation

A 10-fold StratifiedKFold cross-validation was performed on the 90% training set. Table 2 reports the results. The mean cross-validation accuracy was 0.7267 with a standard deviation of 0.0089.

Table 2: 10-Fold Stratified Cross-Validation Results

Fold	Accuracy
1	0.7293
2	0.7196
3	0.7226
4	0.7248
5	0.7393
6	0.7152
7	0.7144
8	0.7296
9	0.7348
10	0.7378
Mean CV Accuracy	0.7267
Std. Dev.	0.0089

4 Conclusions

The decision tree model was then evaluated on the original 10% holdout test set, which had not been used during cross-validation. Table 3 summarizes the model performance of Monte Carlo Cross Validation and 10-Fold Stratified CV and compares them with the holdout test accuracy.

Table 3: Comparison of Model Performance

Method	Mean Accuracy	Std. Dev.	Holdout Test Accuracy
Monte Carlo Cross Validation	0.7231	0.0091	0.7177
10-Fold Stratified CV	0.7267	0.0089	0.7177

The two methods produced very similar estimates of the model’s performance on unseen data. Monte Carlo Cross Validation had a mean out-of-sample accuracy of 0.7231, while 10-fold cross-validation had a slightly higher mean accuracy of 0.7267. The standard deviation was also slightly lower for 10-fold cross-validation (0.0089 versus 0.0091). The final holdout accuracy of 0.7177 was close to both estimates.

In conclusion, 10-fold cross-validation provides a slightly more stable estimate of test performance in this experiment. In terms of computational efficiency, the difference is not pronounced in this experiment because both methods were run only 10 times. However, 10-fold cross-validation is more convenient to implement because the data are automatically divided into multiple folds, whereas Monte Carlo Cross Validation requires explicitly repeating the train-test split and model-fitting process in a loop.

A GitHub Repository

The code and analysis for this report are available in the GitHub repository:

<https://github.com/ronggan01/Machine-Learning-in-Finance-Fall-2026>